University of Central Punjab

Faculty of Information Technology

Artificial Intelligence-LAB

Practice Question for FINAL TERM

Minimax:

1. Implement Minmax algo as per user input like (min max).
2. Find Path for optimal value in reverse order.
3. Break the back tracking at specific Level.
4. Replace the terminal values at specific level.

AlphaBeta:

1. Implement Alpha algo as per user input like (min max).
2. Find Path for optimal value.
3. Break the back tracking at specific Level.
4. Display and count the pruned nodes.
5. Update alpha beta at specific level.

Genetic Algorithm:

1. Implement Genetic Algorithm for given chromosomes.
2. Implement Genetic Algorithm but you are not allowed to use the crossover.
3. Prepare mini problems for Genetic Algorithm **(Use AI tools for those mini problems statement).**
4. Prepare all the past papers especially Genetic Algo questions.
5. Have a better grip on dictionary and loops for better GA problem solving.

Machine Learning:

1. You are supposed to prepare all the libraries.
2. Practice different datasets in order to understand the input features and target variables.
3. Try to implement **Label encoding and one hot encoding in efficient way**.
4. Practice for TF-IDF vectorization and lemmatization for better score in final term.
5. Practice different model training for different test sizes.
6. Visualize the results with best fit technique.
7. Generate confusion matrix for each Model performance.
8. Revise all the theoretical terms like Label encoding, oneHot Encoding, Tokenization and lemmatization with examples.
9. Prepare all the evaluation matrices like Accuracy, FI Score, Recall, Confusion matrix.
10. Compare dataset labels on your own choice.
11. Try to display and compare results before and after preprocessing.
12. Perform model training and testing on user prompts.
13. Test the model on user inputs.
14. Practice how to remove the features having no impact on target value.
15. **TRY TO SOLVE ML PART AT START. OTHERWISE, YOU WILL BE STUCK AND GOES ON DUCK.**

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